Phase-dependent recovery and lifetime of encoded entanglement under collective detector noise

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Abstract

We study how a local preparation phase controls entanglement recovery in a four-qubit encoding whose two detector qubits interact with a common field. All preparations have the same initial reference–register negativity and the same detector marginal. Within a collective secular weak-coupling generator, an exact trace-norm reduction proves an optimal phase at every time in this family. A readout using a Dicke-basis measurement and conditional phase correction retains all outcomes and recovers a two-qubit resource. Its negativity equals a preparation factor times the total variation distance between two evolved sector populations, separating loss of a distinguishable sector label from loss of inter-sector coherence. Weak singlet–triplet mixing produces fast initial relaxation followed by a slow decay. Projection onto the slow sector gives its leading rate and explains when the recovered resource attains the full negativity asymptotically. Applying this law to established accelerated-vacuum and inertial thermal spectra yields a leading lifetime ratio at matched local rates and an observation time maximizing the resource contrast. Common-coupling protection is exact under the stated symmetry. The finite-separation lifetime remains a conditional weak-coupling prediction; its limit order and sensitivity to additional noise are specified. Optimality is confined to the stated preparation family and recovery procedure.

Keywords: Encoded entanglement, Collective noise, Quantum recovery, Metastability, Unruh–DeWitt detectors

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1. Introduction

The usefulness of an entangled register depends on how its correlations are encoded and recovered, not only on the noise acting on its physical constituents. Quantum error correction, noiseless subsystems and decoherence-free subspaces exploit this distinction by placing logical information in degrees of freedom that are less sensitive to a given noise process [1, 2, 3]. A more specific question then arises: can a restricted local preparation improve the entanglement retained by a specified recovery procedure when neither the initial reference entanglement nor the state presented to the environment changes? Answering this question requires separating the amount of initial entanglement from its compatibility with the noise.

We address this question for a four-qubit cluster-state encoding [4]. One qubit serves as a reference, one register qubit remains uncoupled, and two register qubits interact with a field. Recent work reported maximal reference–register negativity when only one detector of a four-qubit cluster state is coupled to the field [5]. That protection can be understood through erasure correctability. When both detectors interact, however, the relevant structure is the singlet–triplet decomposition of their joint noise. Two preparations may then have the same detector marginal but distribute their reference–register coherences differently between these sectors, leading to different recovered resources.

Basis-dependent robustness of cluster states is not itself new. Local rotations have been used to compare equivalent cluster-state representations under independent dephasing [6], and superoperator methods have related four-qubit cluster entanglement to the accuracy of measurement-based operations under independent local noise [7]. The present problem differs in three respects: only the two detectors are noisy; the preparation phase preserves their joint, rather than merely local, marginal state; and the environment generates collective transition channels. The central issue is therefore the exact phase dependence of the encoded resource and of a specified readout, including the slow loss of protection when singlet and triplet sectors begin to mix.

The analysis proceeds in three steps. First, we derive an exact expression for the reference–register negativity under a collective secular weak-coupling generator. The expression reduces the dynamics to four conditional sector populations and one decaying coherence, and proves an all-time phase optimum within the local preparation family. Second, we construct a nonselective

Dicke-basis readout with outcome-dependent phase corrections. Its recovered negativity is a phase factor multiplied by the classical distinguishability of the two evolved sector populations, providing an operational interpretation of the decay. Third, we treat weak sector mixing as a metastability problem in the sense of separated fast and slow relaxation scales [8]. A two-dimensional slow projection determines both the leading lifetime and the phase-dependent amplitude. These are results for the stated family and protocol, not an optimization over arbitrary encodings or recovery maps.

Accelerated detectors provide a controlled application of this resource law. The local response in accelerated vacuum has the Unruh temperature [9], but two-position correlations differ from those in an inertial thermal bath at the corresponding temperature [10]. We adopt the established detector spectra and population dynamics of Ref. [10] and ask how their spatial correlations affect the encoded-resource lifetime. Matching the local rates isolates this spatial effect and yields a leading decay-rate ratio as well as an analytic observation time maximizing the chosen resource contrast.

Throughout, we distinguish exact common-coupling protection from finite-separation metastability calculated with an effective master equation. Spatial regulation and an ordered limit—weak coupling first, small separation second—specify the conditional microscopic interpretation of the latter [11, 12, 13, 14]. Section 2 defines the encoding and permitted recovery operations, and Section 3 introduces the collective dynamics. Section 4 proves the phase optimum and derives the readout resource. Section 5 obtains its slow lifetime and develops the environmental comparison. Sections 6 and 7 delimit the approximations and summarize the conclusions. Supplementary Material Sections S1–S5 provide the encoding identities, field spectra, detailed trace-norm calculation, slow projection and stability analysis; Section S6 records the numerical checks.

2. Encoding and allowed recovery

2.1. Preparation and the quantities held fixed

Let A denote the reference and BCD the register, with only C and D coupled to the environment. After the interaction, joint quantum operations within BCD are allowed; this operational assumption is essential to the recovery statements below. Free phases of spectator qubits are absorbed into the interaction picture and the recovery gates. We set $\hbar = c = k_B = 1$

and quantify entanglement across $A|BCD$ by the normalized negativity

$$\mathcal{N}(\rho) = \|\rho^{T_A}\|_1 - 1, \qquad \|Q\|_1 = \mathrm{tr}\sqrt{Q^\dagger Q}. \tag{1}$$

Here T_A is partial transpose on A, tr is the trace, and $Q^\dagger$ denotes the adjoint of an operator Q. For a normalized vector v, write $P_v = |v\rangle\langle v|$. The normalization is twice the convention of Ref. [15]; its maximum across $A|BCD$ is one. It measures the specified bipartition, not genuine four-party entanglement.

Apply a Pauli flip X_D and $R_C(\theta) = \mathrm{diag}(1, e^{i\theta})$ to $|\mathrm{CL}_4\rangle = (|0000\rangle + |0011\rangle + |1100\rangle - |1111\rangle)/2$. The resulting family is

$$|\psi_\theta\rangle = \frac{|00\rangle_{AB}(|01\rangle + e^{i\theta}|10\rangle)_{CD} + |11\rangle_{AB}(|01\rangle - e^{i\theta}|10\rangle)_{CD}}{2}. \tag{2}$$

To see which information is encoded, define the two orthogonal detector states

$$\begin{aligned} |\zeta_\pm\rangle &= \frac{|01\rangle \pm e^{i\theta}|10\rangle}{\sqrt{2}}, \\ |\psi_\theta\rangle &= \frac{|0\rangle_A|0\rangle_B|\zeta_+\rangle_{CD} + |1\rangle_A|1\rangle_B|\zeta_-\rangle_{CD}}{\sqrt{2}}. \end{aligned} \tag{3}$$

The two register vectors are orthonormal, so this is a Schmidt decomposition with equal coefficients. Thus $\rho_A = I_A/2$ and $\mathcal{N}(0) = 1$ for every phase. Tracing out AB also gives the same detector marginal,

$$\rho_{CD}(0) = \tfrac{1}{2}(P_t + P_s), \quad |t\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \quad |s\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}, \quad P_j = |j\rangle\langle j|. \tag{4}$$

Define $|g\rangle = |00\rangle$, $|e\rangle = |11\rangle$ and $P_T = P_g + P_t + P_e = I - P_s$, where I is the identity on CD. Thus θ controls neither the initial negativity nor the detector populations. Instead, it controls how the reference–register coherence is distributed between the symmetric and antisymmetric detector sectors. This separation excludes a trivial explanation of the phase effect in terms of different initial resources or different local noise strengths.

The initial AB pair is not already a Bell pair: orthogonality of $\zeta_\pm$ gives $\rho_{AB}(0) = (P_{00} + P_{11})/2$, which is separable. The entanglement is stored between A and the full register. The operational sequence is preparation, evolution of CD with A, B uncoupled, and recovery after the detector–field interaction is switched off. The readout in Section 4.2 measures CD jointly and applies a conditional phase gate to B before discarding the detectors.

2.2. Single-detector noise and sector preservation

Before addressing joint noise, it is useful to isolate why the original encoding is protected against one-detector noise. The cluster encoding satisfies the erasure form of the Knill–Laflamme condition for loss of either C or D [1]. For example, discarding D leaves an equal mixture of $P^{AB}_{\Phi^+} \otimes P^C_0$ and $P^{AB}_{\Phi^-} \otimes P^C_1$, where $|\Phi^\pm\rangle = (|00\rangle \pm |11\rangle)/\sqrt{2}$. The orthogonal C flags preserve negativity one in the reduced state. Because an arbitrary trace-preserving channel on D does not change this reduced state, monotonicity under discarding D and the qubit-reference upper bound imply that the full output negativity also remains one. Exchanging the detectors and applying the local preparation unitaries give the other cases. This argument explains single-detector protection without invoking relativistic physics; Supplementary Material S1 gives the corresponding code conditions.

For joint noise, consider a completely positive trace-preserving channel $\mathcal{E}$ admitting Kraus operators such that

$$\mathcal{E}(\rho) = \sum_\mu K_\mu \rho K_\mu^\dagger, \qquad [K_\mu, P_s] = 0, \qquad \sum_\mu K_\mu^\dagger K_\mu = I. \tag{5}$$

Each Kraus operator is block diagonal with respect to the singlet–triplet decomposition. Consequently, $\mathcal{E}(P_t)$ stays in the triplet sector and $\mathcal{E}(P_s) = P_s$. Exchange covariance alone does not imply this property: a covariant channel can still transfer population between sectors. Supplementary Material S1 gives an explicit counterexample. Let id_{AB} denote the identity channel on AB.

Proposition 1 (Sector protection)**.** *For a channel satisfying Eq.* (5)*,*

$$\mathcal{N}\Big[(\mathrm{id}_{AB} \otimes \mathcal{E})(|\psi_\theta\rangle\langle\psi_\theta|)\Big] \geq |\sin\theta|. \tag{6}$$

Proof. Resolve $|\psi_\theta\rangle$ as $(|\phi_T\rangle|t\rangle + |\phi_s\rangle|s\rangle)/\sqrt{2}$, where

$$|\phi_{T,s}\rangle = \tfrac{1}{2}[(1 \pm e^{i\theta})|00\rangle + (1 \mp e^{i\theta})|11\rangle].$$

For a normalized state $x|00\rangle + y|11\rangle$, the present negativity is $2|xy|$. Hence each $\phi_{T,s}$ has negativity $|1 - e^{2i\theta}|/2 = |\sin\theta|$. Applying the channel and then a nonselective measurement of P_T, P_s gives

$$\rho_{\mathrm{flag}} = \tfrac{1}{2} P_{\phi_T} \otimes \mathcal{E}(P_t) + \tfrac{1}{2} P_{\phi_s} \otimes P_s. \tag{7}$$

The detector factors have orthogonal supports and unit trace. The partially transposed trace norm therefore adds over the two flag blocks, giving negativity $|\sin\theta|$. Since the register-side measurement cannot increase negativity, the unmeasured output obeys the same lower bound. □

The two extreme phases explain the bound. At $\theta = 0$, the conditional states are $\phi_T = |00\rangle$ and $\phi_s = |11\rangle$; the reference–register entanglement depends on coherence between the detector flags. At $\theta = \pi/2$, each conditional AB state is maximally entangled. Erasing coherence between the flags can then leave a full Bell pair recoverable, provided the flags themselves remain distinct.

At $\theta = \pi/2$, a fixed register unitary recovers a Bell pair:

$$V = \mathrm{diag}(1, i)_B \otimes P_T + \mathrm{diag}(1, -i)_B \otimes P_s, \tag{8}$$

$$(I_A \otimes V)(\mathrm{id}_{AB} \otimes \mathcal{E})(P_{\psi_{\pi/2}})(I_A \otimes V^\dagger) = P_{\Phi^+} \otimes \mathcal{E}(P_\xi), \tag{9}$$

$$|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2},$$

$$|\xi\rangle = (e^{i\pi/4}|t\rangle + e^{-i\pi/4}|s\rangle)/\sqrt{2}.$$

To verify Eq. (9), note that at $\theta = \pi/2$ the two controlled gates map $|\phi_T\rangle$ and $|\phi_s\rangle$ to $e^{i\pi/4}|\Phi^+\rangle$ and $e^{-i\pi/4}|\Phi^+\rangle$, respectively. Hence V factorizes the prepared state into a Bell pair on AB and $|\xi\rangle$ on CD. Because V commutes with every $I_B \otimes K_\mu$, the same factorization holds after the channel. The optimal phase therefore makes both conditional AB states maximally entangled, even if coherence between the detector sectors has been lost.

This identity also explains the precise scope of the protection claim. The decoder could be applied before the noise, transferring the Bell pair directly to the uncoupled AB subsystem. Equation (9) thus establishes compatibility between this encoding and sector-preserving noise; it does not demonstrate an advantage over unrestricted joint preprocessing. Nor is the coherent decoder a protocol using only local operations among four spatially separated parties.

3. Collective detector dynamics

3.1. Collective jumps and relaxation scales

Take identical two-level detectors of gap $\omega > 0$, downward rate $\gamma_\downarrow > 0$ and upward rate $\gamma_\uparrow \geq 0$. Equal gaps allow the two transitions to retain their

cross correlations under secular averaging. For the environments used here, the cross spectrum is real and has the same normalized spatial factor at the positive and negative gap. Weak coupling and secular averaging then give the Gorini–Kossakowski–Sudarshan–Lindblad generator [10, 16, 17]

$$\dot{\rho} = -i[H_{\rm eff}, \rho] + \sum_{\eta=\pm1} (1+\eta f)\Big(\gamma_\downarrow \mathcal{D}[J_\eta]\rho + \gamma_\uparrow \mathcal{D}[J_\eta^\dagger]\rho\Big), \tag{10}$$
$$J_\pm = (\sigma_C^- \pm \sigma_D^-)/\sqrt{2}, \qquad \mathcal{D}[J]\rho = J\rho J^\dagger - \tfrac{1}{2}\{J^\dagger J, \rho\}.$$

Denote the right-hand side of Eq. (10) by $\mathcal{L}(\rho)$. The dot denotes $d/d\tau$ in proper time, $\sigma_j^- = |0\rangle_j\langle 1|$, and braces denote the anticommutator. Positivity of the spectral matrix requires $|f| \le 1$. The factors $1+f$ and $1-f$ weight the symmetric and antisymmetric ladders, whose nonzero downward actions are

$$J_+|e\rangle = |t\rangle, \quad J_+|t\rangle = |g\rangle, \qquad J_-|e\rangle = |s\rangle, \quad J_-|s\rangle = -|g\rangle. \tag{11}$$

Thus $f = 1$ removes every transition involving the singlet, whereas $|f| < 1$ allows both ladders. At $f = 0$ the dissipator reduces to a sum of local detector dissipators. The renormalized Hamiltonian $H_{\rm eff}$ is assumed diagonal in the Dicke basis (g, t, e, s) [18]. It does not affect the population equations; its remaining coherence phase cancels in the negativity for this encoding. Define

$$q = \gamma_\uparrow/\gamma_\downarrow, \quad \Gamma = \gamma_\downarrow + \gamma_\uparrow, \quad z = \Gamma\tau, \quad r_\pm = \frac{1 \pm f}{1+q}. \tag{12}$$

Here q is the excitation-to-decay ratio, Γ is the local relaxation rate, and z is dimensionless time. The dimensionless downward rates of the two ladders are $r_\pm$; the corresponding upward rates are $qr_\pm$. We use $0 < q < 1$ for the thermal slow-limit analysis and treat $q = 0$ separately. The microscopic scope of this effective generator is specified in Section 6.

3.2. Two population histories and their difference

Define $p_j = \langle j|\rho|j\rangle$ for $j = g, t, e, s$. The population column vector $\boldsymbol{p} = (p_g, p_t, p_e, p_s)^{\mathsf{T}}$ uses ordinary transpose T. Its evolution is $d\boldsymbol{p}/dz = M\boldsymbol{p}$, with the established rate matrix

$$M = \begin{pmatrix} -q(r_+ + r_-) & r_+ & 0 & r_- \\ qr_+ & -(1+q)r_+ & r_+ & 0 \\ 0 & qr_+ & -(r_+ + r_-) & qr_- \\ qr_- & 0 & r_- & -(1+q)r_- \end{pmatrix}. \tag{13}$$

For example, the physical triplet balance equation is

$$\frac{dp_t}{d\tau} = (1+f)\Big[\gamma_\uparrow p_g + \gamma_\downarrow p_e - \Gamma p_t\Big]. \tag{14}$$

Dividing by Γ yields the second row of M; the other rows follow from Eq. (11). The off-diagonal matrix entries are nonnegative rates and every column sums to zero, as required for conservation of probability.

Let $\mathcal{E}_z$ be the detector channel at time z. We compare two auxiliary preparations, P_t and P_s, and define $p_j^{(t)} = \langle j|\mathcal{E}_z(P_t)|j\rangle$, $p_j^{(s)} = \langle j|\mathcal{E}_z(P_s)|j\rangle$, and $d_j = p_j^{(t)} - p_j^{(s)}$. These are two alternative population histories, rather than two components measured simultaneously in one run. Each history is a normalized probability distribution. Their difference obeys the same linear equation and has initial value $(0,1,0,-1)^{\mathsf{T}}$, giving

$$\boldsymbol{d}(z) = e^{Mz}(0,1,0,-1)^{\mathsf{T}}. \tag{15}$$

The entries of $\boldsymbol{d}$ may be negative, but $\sum_j d_j = 0$. For $q > 0$ and $|f| < 1$, the population graph is connected. Its unique stationary distribution is

$$\boldsymbol{p}_\infty = \frac{(1,q,q^2,q)^{\mathsf{T}}}{(1+q)^2}, \tag{16}$$

as verified by $M\boldsymbol{p}_\infty = 0$. It is the Dicke-basis population of the bare detectors' product Gibbs state, with local excitation-to-ground weight ratio q. Both auxiliary histories approach it, so $\boldsymbol{d} \to 0$. At $f = 1$, the singlet is disconnected from the triplet, and the two histories remain distinguishable. This change in connectivity is the origin of the slow leakage studied below.

4. Exact phase optimum and a specified readout

4.1. Reduction to one detector operator

The relevant coherence is between the $AB = 00$ and $AB = 11$ branches in Eq. (3). Its evolution is determined by the cross operator

$$W_\theta(z) = \mathcal{E}_z(|\zeta_+\rangle\langle\zeta_-|). \tag{17}$$

This operator is not a density matrix: its trace is zero and it need not be Hermitian. Define the phase parameters and squared coherence modulus by

$$h = |\sin\theta|, \quad \alpha = \tfrac{1}{2}\sin\theta, \quad b = \frac{1+\cos\theta}{2}, \quad c = \frac{1-\cos\theta}{2}, \quad E = e^{-2z}. \tag{18}$$

They obey $b + c = 1$, $bc = \alpha^2$ and $h = 2|\alpha|$.

Proposition 2 (Exact negativity and phase optimum)**.** *For the encoding in Eq.* (2) *and the generator in Eq.* (10)*,*

$$\mathcal{N}_\theta(z) = \|W_\theta(z)\|_1, \tag{19}$$

$$\mathcal{N}_\theta(z) = \frac{h}{2}(|d_g| + |d_e|) + \sqrt{E + h^2 K(z)}, \tag{20}$$

$$K(z) = \tfrac{1}{4}\Big[d_t^2 + d_s^2 - 2E + 2|d_t d_s + E|\Big] \geq 0. \tag{21}$$

Consequently $e^{-z} = \mathcal{N}_0(z) \leq \mathcal{N}_\theta(z) \leq \mathcal{N}_{\pi/2}(z)$ *at every finite time. The maximizing phase is* $\pi/2$ *modulo* π*, with possible degeneracy.*

Proof. Write $|\psi_\theta\rangle = (|00\rangle|\zeta_+\rangle + |11\rangle|\zeta_-\rangle)/\sqrt{2}$. Partial transpose on A puts the cross terms into an independent $AB = 01, 10$ block,

$$\tfrac{1}{2}\begin{pmatrix} 0 & W_\theta^\dagger \\ W_\theta & 0 \end{pmatrix}.$$

Its eigenvalues are plus and minus half the singular values of W_θ. The remaining two blocks are positive and each has trace $1/2$. This proves Eq. (19).

In the Dicke basis, the initial cross operator separates into a population difference and two coherences,

$$|\zeta_+\rangle\langle\zeta_-| = i\alpha(P_t - P_s) + b|t\rangle\langle s| + c|s\rangle\langle t|. \tag{22}$$

The population part evolves to $i\alpha \sum_j d_j P_j$. For $|t\rangle\langle s|$, each jump-gain term vanishes because each collective jump annihilates either t or s. The downward anticommutators contribute $-\gamma_\downarrow|t\rangle\langle s|$ and the upward ones contribute $-\gamma_\uparrow|t\rangle\langle s|$. Let E_t, E_s be the corresponding eigenenergies of H_{eff} and put $\Delta E = E_t - E_s$. Then

$$\begin{aligned} \mathcal{L}(|t\rangle\langle s|) &= -(\Gamma + i\Delta E)|t\rangle\langle s|, \\ \mathcal{E}_z(|t\rangle\langle s|) &= e^{-z-i\varphi}|t\rangle\langle s|, \qquad \varphi = \Delta E\, z/\Gamma. \end{aligned} \tag{23}$$

The conjugate coherence acquires the opposite phase. Thus, in the order (g, t, e, s),

$$W_\theta = \begin{pmatrix} i\alpha d_g & 0 & 0 & 0 \\ 0 & i\alpha d_t & 0 & be^{-z-i\varphi} \\ 0 & 0 & i\alpha d_e & 0 \\ 0 & ce^{-z+i\varphi} & 0 & i\alpha d_s \end{pmatrix}. \tag{24}$$

Reordering the basis to (g, e, t, s) exhibits two scalar blocks and the 2×2 block Y on t, s. The scalar blocks contribute $|\alpha|(|d_g|+|d_e|)$ to the trace norm. For its two singular values, the sum of squares is $\mathrm{tr}(YY^\dagger)$ and the product is $|\det Y|$. Therefore $\|Y\|_1^2 = \mathrm{tr}(YY^\dagger) + 2|\det Y|$, with

$$\mathrm{tr}(YY^\dagger) = \alpha^2(d_t^2 + d_s^2) + (b^2 + c^2)E, \qquad \det Y = -\alpha^2(d_t d_s + E).$$

Using $b^2+c^2 = 1-h^2/2$ proves Eqs. (20)–(21); the Hamiltonian phase cancels.

If $d_t d_s + E \geq 0$, then $K = (d_t + d_s)^2/4$. Otherwise $(d_t - d_s)^2 \geq -4d_t d_s > 4E$ gives $K = [(d_t - d_s)^2 - 4E]/4 \geq 0$. For finite z, $E > 0$ and

$$\frac{\partial \mathcal{N}_\theta}{\partial h} = \frac{|d_g| + |d_e|}{2} + \frac{hK}{\sqrt{E + h^2 K}} \geq 0. \tag{25}$$

Hence $h = 1$ maximizes the negativity and $h = 0$ gives e^{-z}. This proves both bounds. □

Equation (20) makes the origin of the optimum explicit. The phase enters only through $h = |\sin\theta|$; the first term contains the g, e sector-population differences, while the square root combines the t, s population differences with the surviving inter-sector coherence. Since $K \geq 0$ and is independent of θ, increasing h can never decrease the negativity. Thus the phase optimum follows from the structure of the evolved cross operator, rather than from a numerical search or a change in the initial resource.

Two limiting checks help interpret this result. At $f = 0$, Eq. (13) gives $d_g = d_e = 0$ and $d_t = -d_s = e^{-z}$, so every phase has the same negativity $\mathcal{N} = e^{-z}$. Spatially correlated dissipative transitions are therefore essential for this phase dependence within the stated model. At every finite z, the positive term E in Eq. (20) ensures $\mathcal{N}_\theta > 0$. This statement is specific to the bipartition $A|BCD$ and normalized negativity; it does not exclude sudden death for another bipartition or entanglement measure.

4.2. Measurement, correction and the recovered pair

The readout has three steps: measure CD in the Dicke basis, apply an outcome-dependent phase gate to B, and average over all outcomes. The measurement basis is fixed. The phase gates are calibrated from the known preparation and dynamics; this is not a single recovery map assumed to work without knowledge of the channel. Distinguishing t from s requires a joint detector measurement, consistent with the operations permitted in Section 2.

For outcome $j \in \{g,t,e,s\}$, the unnormalized conditional state is $\rho_{AB}^{(j)} = \langle j|\rho_{ABCD}(z)|j\rangle$. In its supported basis $(00,11)$ it is

$$\begin{aligned} \rho_{AB}^{(j)} &= \frac{1}{2}\begin{pmatrix} bp_j^{(t)} + cp_j^{(s)} & i\alpha d_j \\ -i\alpha d_j & cp_j^{(t)} + bp_j^{(s)} \end{pmatrix}, \\ \pi_j &= \operatorname{tr}\rho_{AB}^{(j)} = \frac{p_j^{(t)} + p_j^{(s)}}{2}. \end{aligned} \tag{26}$$

This follows because the Dicke populations of P_{ζ_+} are initially (b,c) on (t,s), those of P_{ζ_-} are (c,b), and their cross operator has diagonal entry $i\alpha d_j$. The outcome probabilities π_j are independent of θ, as expected from the common detector marginal.

Choose

$$U_B^{(j)} = \operatorname{diag}(1, e^{i\vartheta_j}), \qquad \vartheta_j = \arg(i\alpha d_j), \tag{27}$$

with $\vartheta_j = 0$ if $\alpha d_j = 0$. Conjugation multiplies the $00,11$ entry by $e^{-i\vartheta_j}$, making it real and nonnegative. Summing the corrected, unnormalized branches retains all probabilities, rather than postselecting an outcome. The diagonal entries sum to $1/2$ each and the off-diagonal entry becomes $h\sum_j |d_j|/4$. Therefore

$$\mathcal{N}_{\mathrm{rec},\theta}(z) = \frac{h}{2}\sum_{j=g,t,e,s}|d_j(z)| = h\,\mathrm{TV}(\boldsymbol{p}^{(t)}, \boldsymbol{p}^{(s)}) \le \mathcal{N}_\theta(z), \tag{28}$$

$$\rho_{AB}^{\mathrm{rec}} = \tfrac{1}{2}(P_{00} + P_{11}) + \tfrac{\mathcal{N}_{\mathrm{rec},\theta}}{2}(|00\rangle\langle 11| + |11\rangle\langle 00|). \tag{29}$$

Its partial transpose has eigenvalues $1/2$ (twice) and $\pm\mathcal{N}_{\mathrm{rec},\theta}/2$, verifying the negativity in Eq. (28). The inequality follows because measurement, correction and discarding CD all act on the register side.

Here $\mathrm{TV}(\boldsymbol{p},\boldsymbol{q}) = \frac{1}{2}\sum_j |p_j - q_j|$ is total variation distance. It quantifies the remaining distinguishability of the two auxiliary sector preparations. It is not a probability difference between two phases: the actual outcome probabilities are the π_j in Eq. (26). Since $T(\Delta z) = e^{M\Delta z}$ is a stochastic matrix for $\Delta z \ge 0$,

$$\begin{aligned} \|\boldsymbol{d}(z+\Delta z)\|_1 &= \|T(\Delta z)\boldsymbol{d}(z)\|_1 \\ &\le \sum_{i,j} T_{ij}(\Delta z)|d_j(z)| = \|\boldsymbol{d}(z)\|_1. \end{aligned} \tag{30}$$

Thus the specified recovered negativity cannot revive within this time-independent Markov model. At $z = 0$ it equals h, whereas the full negativity

is one: the measurement can discard useful inter-sector coherence. The subscript "rec" always refers to this calibrated procedure, not to optimization over every recovery map. Supplementary Fig. S1 illustrates the distinction during fast and slow evolution.

For a task interpretation, put $r = \mathcal{N}_{\mathrm{rec},\theta}$. Equation (29) is the Bell mixture $(1+r)P_{\Phi^+}/2 + (1-r)P_{\Phi^-}/2$. Standard Bell-measurement teleportation then gives the dephasing channel [19]

$$\mathcal{T}_r(\rho_{\mathrm{in}}) = \frac{1+r}{2}\rho_{\mathrm{in}} + \frac{1-r}{2}\sigma_z\rho_{\mathrm{in}}\sigma_z, \qquad \sigma_z = \mathrm{diag}(1,-1), \tag{31}$$

where ρ_{in} is the unknown qubit input. For a pure input with Bloch component n_z, its fidelity is $(1+r)/2 + (1-r)n_z^2/2$. The uniform Bloch-sphere average has $\langle n_z^2\rangle = 1/3$, giving the known relation

$$\overline{F} = \frac{2+r}{3} = \frac{2+\mathcal{N}_{\mathrm{rec},\theta}}{3}. \tag{32}$$

It exceeds the classical threshold $2/3$ whenever $r > 0$. The excess fidelity is $r/3$, so it inherits the same slow decay as the recovered resource. No new teleportation protocol is introduced.

The physical mechanism can now be stated in two stages. First, the collective channel distinguishes symmetric from antisymmetric detector amplitudes. The relative preparation phase changes their correlations with AB without changing $\rho_{CD}(0)$, which explains why the phase matters only when the two detector transitions are spatially correlated. Second, loss of t–s coherence and loss of sector-population distinguishability are distinct processes. Conditional AB entanglement can survive the former as long as the evolved triplet and singlet populations remain distinguishable; sector mixing erodes the latter and hence suppresses the recovered resource. The independent-noise case $f = 0$ confirms the first point, while the slow-sector analysis below quantifies the second.

4.3. Boundary cases and consistency checks

At $q = 0$ there are no upward transitions. The initial difference has $d_e = 0$, and the two one-excitation populations decay independently:

$$d_t = e^{-(1+f)z}, \quad d_s = -e^{-(1-f)z}, \quad d_e = 0, \quad d_g = -d_t - d_s. \tag{33}$$

Substituting these expressions in Eq. (20) gives

$$\mathcal{N}_\theta = e^{-z}\left[h|\sinh(fz)| + \sqrt{1+h^2\sinh^2(fz)}\right], \qquad \mathcal{N}_{\pi/2} = e^{-(1-|f|)z}. \tag{34}$$

Table 1: Consistency checks within the effective detector generator. The notation $z \to \infty$ denotes the long-time limit taken at the indicated fixed parameters.

Condition	$\mathcal{N}_\theta$	$\mathcal{N}_{\mathrm{rec},\theta}$
$z = 0$	1	h
$f = 0$, any z	e^{-z}	he^{-z}
$f = 1$, $h = 1$, any z	1	1
$f = 1$, $z \to \infty$	h	h
$\lvert f\rvert < 1$, $q > 0$, $z \to \infty$	0	0

For the optimal phase, the zero-temperature result is therefore a single exponential; near $f = 1$ its rate is $(1 - f)\Gamma$. This is an exact boundary solution, separate from the fixed-$q > 0$ expansion below. Table 1 collects further limits of the full and specified recovered resources.

5. Metastable lifetime and an environmental comparison

5.1. Weak sector mixing

Near common coupling, write $\epsilon = 1 - f \ll 1$, so that ϵ measures the strength of singlet–triplet mixing. Define $Z = 1 + q + q^2$ and

$$\mathcal{C}(q) = 1 + \frac{q}{Z} = \frac{(1+q)^2}{Z}. \tag{35}$$

At $\epsilon = 0$, population dynamics within the triplet sector relax to $\boldsymbol{r}_T = (1, q, q^2, 0)^{\mathsf{T}}/Z$, while the singlet vector $\boldsymbol{r}_s = (0, 0, 0, 1)^{\mathsf{T}}$ is stationary. For small nonzero ϵ, rapid triplet relaxation is followed by slow leakage between the two sectors. Projecting onto the span of $\boldsymbol{r}_T$ and $\boldsymbol{r}_s$ isolates this slow process.

Proposition 3 (Slow resource law)**.** *At fixed $0 < q < 1$, the weakly mixed generator has a physical slow decay rate*

$$\kappa = \epsilon\Gamma\mathcal{C}(q) + O(\epsilon^2\Gamma). \tag{36}$$

At fixed slow time $u = \epsilon z > 0$,

$$\begin{aligned} \boldsymbol{d}(u/\epsilon) &\longrightarrow e^{-\mathcal{C}(q)u}(\boldsymbol{r}_T - \boldsymbol{r}_s), \\ \mathcal{N}_\theta(u/\epsilon),\ \mathcal{N}_{\mathrm{rec},\theta}(u/\epsilon) &\longrightarrow h\, e^{-\mathcal{C}(q)u}. \end{aligned} \tag{37}$$

Proof. The rates obey $r_+ = (2-\epsilon)/(1+q)$ and $r_- = \epsilon/(1+q)$, so Eq. (13) is exactly $M = M_0 + \epsilon M_1$, with $M_0 = M|_{f=1}$ and

$$M_1 = \frac{1}{1+q}\begin{pmatrix} 0 & -1 & 0 & 1 \\ -q & 1+q & -1 & 0 \\ 0 & -q & 0 & q \\ q & 0 & 1 & -(1+q) \end{pmatrix}. \tag{38}$$

The left zero vectors $\boldsymbol{\ell}_T = (1,1,1,0)$ and $\boldsymbol{\ell}_s = (0,0,0,1)$ obey $\boldsymbol{\ell}_i \boldsymbol{r}_j = \delta_{ij}$. Projection gives

$$[\boldsymbol{\ell}_i M_1 \boldsymbol{r}_j]_{i,j=T,s} = \begin{pmatrix} -q/Z & 1 \\ q/Z & -1 \end{pmatrix}. \tag{39}$$

The singlet-row entries are $\boldsymbol{\ell}_s M_1 \boldsymbol{r}_T = (q+q^2)/[(1+q)Z] = q/Z$ and $\boldsymbol{\ell}_s M_1 \boldsymbol{r}_s = -1$. Column conservation fixes the other row. The projected matrix has determinant zero and trace $-\mathcal{C}(q)$, so its eigenvalues are zero and $-\mathcal{C}(q)$. The other eigenvalues of M_0 are $-2 \pm 2\sqrt{q}/(1+q)$ and remain separated from zero. Finite-dimensional perturbation about this isolated two-dimensional subspace gives one stationary eigenvalue and one eigenvalue $-\epsilon\mathcal{C}(q) + O(\epsilon^2)$; multiplication by Γ gives Eq. (36).

The initial difference has sector coordinates $(1,-1)^{\mathsf{T}}$ and zero total mass. Its slow projection is $\boldsymbol{r}_T - \boldsymbol{r}_s$. The coordinate vector $(1,-1)^{\mathsf{T}}$ is the eigenvector of Eq. (39) with eigenvalue $-\mathcal{C}(q)$, while the stationary coordinate vector has nonzero total mass. Thus no stationary component survives in the population difference. At $z = u/\epsilon$, the fast population modes and the coherence e^{-z} vanish, while the difference mode tends to $e^{-\mathcal{C}(q)u}$. Substitution into Eqs. (20) and (28) proves Eq. (37), since the two limiting population vectors have disjoint supports and $\|\boldsymbol{r}_T - \boldsymbol{r}_s\|_1 = 2$. Errors are $O(\epsilon)$ on compact positive intervals of u, with an additional exponentially small fast contribution. Supplementary Material S4 gives M_1 explicitly. □

The two entries in $\mathcal{C}(q) = 1 + q/Z$ are the singlet-to-triplet escape rate and the reverse rate after fast triplet equilibration. Define the slow relaxation time by

$$\tau_{\mathrm{slow}} = \kappa^{-1} = \frac{1 + O(\epsilon)}{\epsilon\Gamma\mathcal{C}(q)}. \tag{40}$$

For $h > 0$, this is the $1/e$ time of the slow envelope relative to its amplitude h. It is not generally the time at which the full negativity, initially one, first falls to $1/e$. In particular, $h = 0$ has no slow recovered resource: $\mathcal{N}_{\mathrm{rec},0} =$

0 and $\mathcal{N}_0 = e^{-\Gamma\tau}$ exactly. On fixed positive slow times, the inter-sector coherence is exponentially small, so the surviving negativity is determined by the population difference and the specified readout attains it to leading order.

The slow limit must be distinguished from the initial dynamics. When $h < 1$ and mixing tends to zero, the full negativity first falls from one towards the plateau amplitude h on the fast scale $z = O(1)$. This transient occupies $u = O(\epsilon)$ and collapses to an initial boundary layer on the slow scale. Convergence of the full negativity is therefore not uniform down to $u = 0$. Even at the optimal phase, finite separation gives $\mathcal{N}_{\pi/2}(z) = 1 - \epsilon z + O(z^2)$ for $f \geq 0$, as follows by inserting $\boldsymbol{d} = \boldsymbol{d}(0) + Mz\boldsymbol{d}(0) + O(z^2)$ into Eq. (20). Thus the metastable plateau is long lived but not exactly frozen. Figure 1 shows both the phase dependence under exact collective coupling and the convergence toward the slow law as mixing decreases.

For $h > 0$, the long-time and collective limits consequently do not commute within the effective dynamics:

$$\lim_{\epsilon\to 0^+}\lim_{z\to\infty}\mathcal{N}_\theta(z;\epsilon) = 0, \qquad \lim_{z\to\infty}\lim_{\epsilon\to 0^+}\mathcal{N}_\theta(z;\epsilon) = h. \tag{41}$$

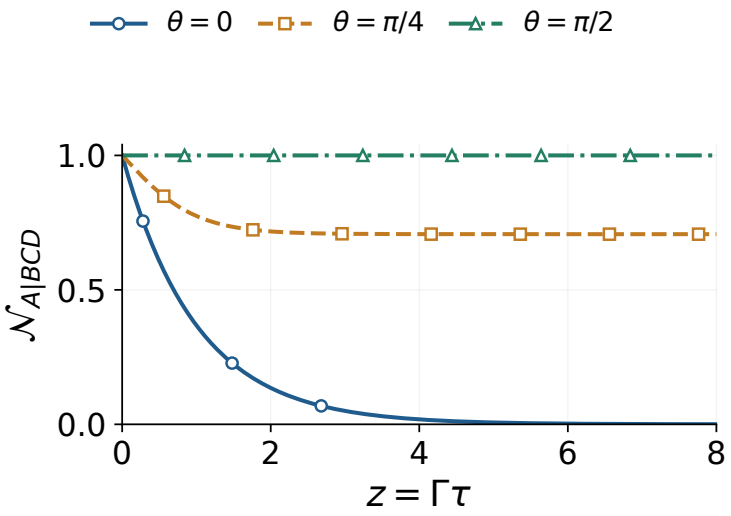
At every fixed nonzero mixing, the sector label is eventually lost; at exact collective coupling it remains available indefinitely. Metastability describes the long crossover between these two behaviors, not an additional stationary entangled state at finite separation.

5.2. Matched local rates in accelerated vacuum and a thermal bath

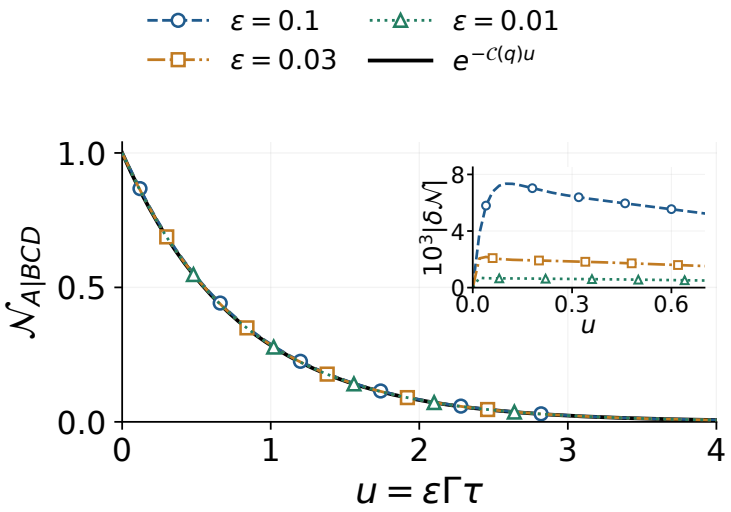
To isolate the effect of spatial correlations, consider two environments with identical local detector response. The first consists of identical detectors with uniform acceleration $a > 0$ and transverse separation L, coupled with dimensionless strength λ to a massless scalar field in four-dimensional Minkowski vacuum. The second consists of inertial detectors at the same separation in a thermal bath of inverse temperature $2\pi/a$. In both cases, transition rates are λ^2 times the field spectra at $\pm\omega$. The point-profile spectra, rederived in Supplementary Material S2 following Ref. [10], give

$$q = e^{-2\pi\omega/a}, \qquad \gamma_\downarrow = \frac{\lambda^2\omega}{2\pi(1-q)}, \qquad \gamma_\uparrow = q\gamma_\downarrow, \tag{42}$$

$$f_{\rm acc} = \frac{\sin[(2\omega/a)\,\mathrm{arsinh}(aL/2)]}{\omega L\sqrt{1+a^2L^2/4}}, \qquad f_{\rm th} = \mathrm{sinc}(\omega L). \tag{43}$$



(a) Exact collective coupling.



(b) Weak sector mixing.

Figure 1: Phase dependence and slow decay at $q = 0.4$. (a) At $f = 1$, the optimal phase retains maximal $A|BCD$ negativity, whereas other phases relax towards $|\sin\theta|$. (b) At $\theta = \pi/2$ and $f = 1 - \epsilon$, the exact generator approaches $e^{-\mathcal{C}(q)u}$ on the slow scale $u = \epsilon\Gamma\tau$. The inset shows the actual absolute errors multiplied by 10^3; curves and markers use unshifted data.

Here $\operatorname{sinc} x = \sin x/x$ with $\operatorname{sinc} 0 = 1$, and arsinh is the inverse hyperbolic sine. The local rates agree, but the spatial factors do not. The quadratic small-separation expansion requires both $\omega L \ll 1$ and $aL \ll 1$. Expanding the known factors at fixed a, ω gives

$$\epsilon_{\rm acc} = \frac{\omega^2 + a^2}{6} L^2 + O(L^4), \qquad \epsilon_{\rm th} = \frac{\omega^2}{6} L^2 + O(L^4),$$
$$\frac{\kappa_{\rm acc}}{\kappa_{\rm th}} \longrightarrow R = 1 + \frac{a^2}{\omega^2}. \tag{44}$$

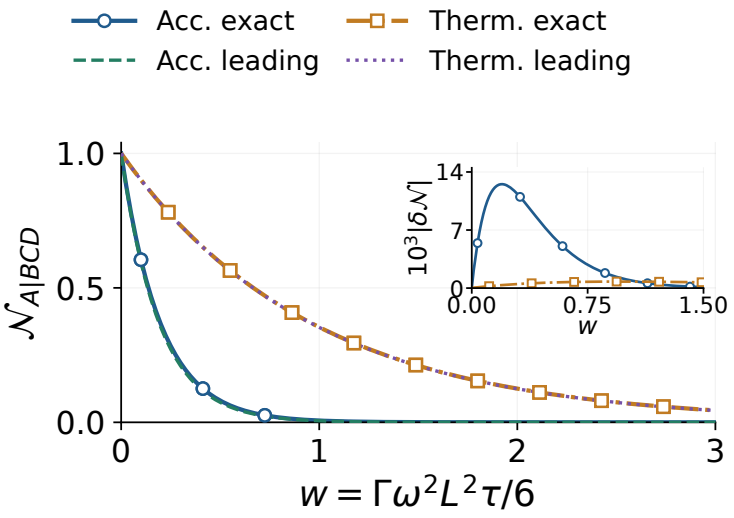
The matching is essential: both environments have the same q and Γ, so their leading slow-decay rates differ only through the small-separation coefficients of ϵ. Consequently, the accelerated resource has the shorter leading lifetime, with $\kappa_{\rm acc}/\kappa_{\rm th} \to R$. The corresponding lifetime ratio is the inverse, $\tau^{\rm acc}_{\rm slow}/\tau^{\rm th}_{\rm slow} \to R^{-1}$. This is not a new calculation of the Unruh spectrum; Eq. (44) translates the established spatial spectra of Ref. [10] into a lifetime statement for the encoded resource.

At the optimal phase and fixed $w = \Gamma\omega^2 L^2 \tau/6 > 0$ as $L \to 0$, Eq. (37) yields $\mathcal{N}_{\rm th} \to e^{-\mathcal{C}w}$ and $\mathcal{N}_{\rm acc} \to e^{-\mathcal{C}Rw}$, with the same limits for the recovered resource. Maximizing their difference gives, for $R > 1$,

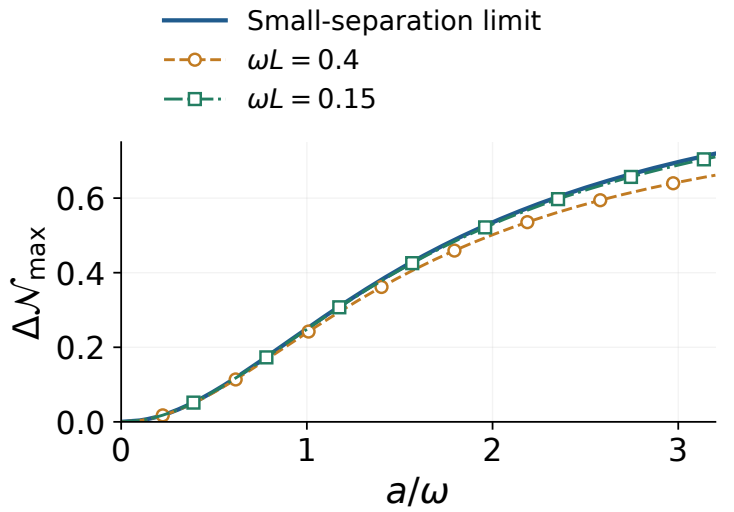
$$w_* = \frac{\ln R}{\mathcal{C}(q)(R-1)}, \qquad \tau_* = \frac{6w_*}{\Gamma\omega^2 L^2}, \tag{45}$$
$$\Delta\mathcal{N}_{\rm max} = (1 - R^{-1}) R^{-1/(R-1)}. \tag{46}$$

For the derivation, $\Delta(w) = e^{-\mathcal{C}w} - e^{-\mathcal{C}Rw}$ has derivative $\mathcal{C}[-e^{-\mathcal{C}w} + Re^{-\mathcal{C}Rw}]$. Its unique positive zero obeys $e^{\mathcal{C}(R-1)w} = R$. Since Δ starts at zero, is initially increasing and returns to zero at infinity, this point is the global maximum. The contrast tends continuously to zero as $a/\omega \to 0$ and $R \to 1$. Acceleration changes the weak sector-mixing rate through the spatial correlation, rather than through the matched local relaxation rate.

These are leading small-separation formulas. The corresponding teleportation-fidelity contrast is $\Delta\mathcal{N}_{\rm max}/3$. This maximizes the stated resource contrast, not the success probability of an optimized environmental discrimination experiment. Figure 2 compares exact-generator dynamics with the environmental limits; sampled values and absolute errors are supplied with the code.



(a) Matched local response.



(b) Optimized negativity contrast.

Figure 2: Environmental application at $\theta = \pi/2$. (a) Exact-generator curves and leading slow laws for $a/\omega = 2$, $\omega L = 0.2$, with common $q = e^{-\pi}$ and local rate Γ. The inset gives absolute errors multiplied by 10^3. (b) Maximum contrast from the full generator at the indicated separations, compared with Eq. (46). "Exact" refers throughout to the effective generator, not to the microscopic detector–field evolution.

5.3. Numerical verification and its scope

Numerical calculations provide two distinct checks. First, direct evolution of the complete four-qubit state agrees with the analytic trace-norm reduction in 800 cases, with maximum absolute error 2.13×10^{-15}. A further 300 boundary and late-time cases, extending to $z = 500$, give a maximum error of 2.60×10^{-13}. Second, convergence tests examine $q \in \{0.01, 0.1, 0.4, 0.9\}$, ϵ between 0.1 and 0.002, $u \in \{0.5, 1, 2\}$ and $\theta \in \{0.37, \pi/2\}$. The endpoint errors decrease toward the slow law as ϵ decreases; at $\epsilon = 0.002$, the largest sampled relative slow-rate error is 1.26×10^{-4} and the largest negativity error is 1.52×10^{-4}. Supplementary Material S6 and the accompanying tables specify the parameter grids and contrast tests. These calculations verify the finite-dimensional formulas and their sampled asymptotic convergence; they neither replace the analytic proofs nor extend the microscopic validity of the effective model.

6. Microscopic scope and sensitivity

6.1. Exact common-coupling protection

The exact protection statement and the effective finite-separation calculation require different levels of justification. For the former, consider smooth, identical detector profiles of radius $r < a^{-1}$, stationary under the common boost and translated transversely by L [12]. The register and field are initially independent, and the detector gaps, couplings and switching functions are identical. At $L = 0$, the smeared field operators coincide. Both the free detector Hamiltonian and the interaction then commute with P_s, as does the full detector–field unitary. Bath matrix elements consequently yield Kraus operators satisfying Eq. (5). Thus the protection bound and decoder in Eqs. (6)–(9) are exact for this common coupling, without invoking a Markov approximation.

6.2. Conditional finite-separation lifetime

For $L > 0$, the lifetime instead relies on the secular weak-coupling description. Its use requires, among other conditions, $\Gamma \ll \omega$, $\Gamma\tau_B \ll 1$ for bath correlation time τ_B, and compatibility with the renormalized Hamiltonian scales [13, 14]. These scale conditions do not by themselves guarantee microscopic accuracy uniformly over a lifetime that diverges as L^{-2}. To state precisely what is established, fix $r, L > 0$ and let $\Gamma_{\lambda,r}$ be the regulated local relaxation rate. Write $f_r(L)$ for the corresponding normalized cross spectrum.

Assume convergence to the effective channel uniformly on bounded intervals of $z = \Gamma_{\lambda,r}\tau$. For fixed geometry and fixed $u > 0$, the value $z = u/[1 - f_r(L)]$ is finite, so this assumption applies when $\lambda \to 0$ at that geometry. Only after that limit do we let $L \to 0$ and use the slow law. This gives the conditional ordered limit

$$\lim_{L\to 0^+} \lim_{\lambda\to 0} \mathcal{N}_\theta^{\mathrm{mic}}\left(\frac{u}{\Gamma_{\lambda,r}[1 - f_r(L)]}\right) = he^{-\mathcal{C}(q)u}, \qquad u > 0. \tag{47}$$

Here the regulated rate ratio is held fixed. Reflection symmetry and a finite fourth spectral moment give $1 - f_r(L) = c_r L^2 + O(L^4)$, where c_r is half the second transverse-momentum moment of the normalized spectrum at frequency ω (Supplementary Material S5). Recovering the point-profile coefficients in Eq. (44) additionally requires convergence of the corresponding on-shell second moments as $r \to 0$. The result therefore does not cover a simultaneous limit $L = L(\lambda)$, nor does it supply a fixed-coupling bound on microscopic memory errors over arbitrarily growing times.

6.3. Sensitivity to additional perturbations

Long lifetimes also amplify the effect of small imperfections. To quantify this sensitivity, let $\mathcal{L}$ be the generator in Eq. (10) and $\widetilde{\mathcal{L}}$ a perturbed time-independent generator, both producing completely positive trace-preserving channels. The diamond norm is the trace-norm map bound stabilized by an arbitrary reference system. If $\|\widetilde{\mathcal{L}} - \mathcal{L}\|_\diamond \le \delta\Gamma$ with dimensionless $\delta \ge 0$, then

$$|\widetilde{\mathcal{N}}_\theta(\tau) - \mathcal{N}_\theta(\tau)| \le 2\delta\Gamma\tau = \frac{2\delta u}{\epsilon}. \tag{48}$$

To obtain Eq. (48), we use Duhamel's identity to express the channel difference as the time integral of two channels sandwiching $\widetilde{\mathcal{L}} - \mathcal{L}$. Each channel has diamond norm one, yielding a channel bound $\delta\Gamma\tau$; the qubit-reference inequality $|\mathcal{N}(\widetilde{\rho}) - \mathcal{N}(\rho)| \le 2\|\widetilde{\rho} - \rho\|_1$ supplies the remaining factor of two. Supplementary Material S5 gives the complete norm argument. On the slow time $\tau = u/(\epsilon\Gamma)$, the error bound is proportional to δ/ϵ, so $\delta \ll \epsilon$ is sufficient to preserve the leading slow law. A sufficient operational requirement is therefore that additional dephasing or, in a near-degenerate description retaining collective terms, a detector-gap mismatch be small compared with $\epsilon\Gamma$. Comparing imperfections only with Γ does not guarantee a small bound over the long lifetime. For a fixed readout calibrated to the ideal channel,

trace-distance contractivity gives the same bound on its recovered negativity. This norm estimate is conservative: it does not prove that every perturbation comparable with $\epsilon\Gamma$ destroys the resource.

7. Discussion

This work connects local phase preparation, exact entanglement dynamics, operational readout and metastable decay within one four-qubit model. Three conclusions follow. First, the trace-norm reduction proves that $\theta = \pi/2$ is optimal at every time within the preparation family, even though all members have the same initial negativity and detector marginal. The advantage therefore lies in the compatibility of the encoded coherence with collective noise, not in a larger initial resource. Second, the nonselective Dicke readout expresses the recovered negativity through sector-population distinguishability. This separates the operational resource from the rapidly decaying inter-sector coherence. Third, weak sector mixing reduces the long-time dynamics to an effective two-sector process, giving the leading lifetime and showing that the full and recovered resources have the same slow-time limit.

The environmental application isolates the role of spatial correlations by matching the local rates in accelerated vacuum and an inertial thermal bath. The resulting decay-rate ratio and optimal observation time are direct consequences of the resource law applied to established spectra. Single-detector protection, singlet symmetry, the collective generator and the teleportation-fidelity relation themselves have established origins; the contribution here is their explicit connection through the phase-dependent trace norm, specified recovery procedure and slow resource law.

These conclusions have a sharply defined scope. They do not optimize over general encodings or recovery maps, quantify genuine four-party entanglement, or claim a new detector spectrum. Common-coupling protection is exact under the stated symmetry, whereas the finite-separation lifetime is an effective-dynamics prediction with the conditional ordered microscopic interpretation in Eq. (47). Extending the analysis to general recovery optimization or to uniform microscopic error control over the full metastable lifetime would require further work.

Supplementary material

The accompanying Supplementary Material provides proofs, the regulated model and spectral calculation. The reproducibility archive contains

Python sources, figure data and verification results.

Declarations

Funding

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Declaration of competing interest

The author declares no competing financial or non-financial interests.

Data availability

No experimental data were used. The numerical verification outputs and figure data are included in the accompanying reproducibility archive. The figures can be regenerated from the supplied Python sources.

Code availability

The accompanying archive contains the model implementation and checks of the channel identities, boundary cases, asymptotic convergence, conditional measurement states and the diagnostic exchange-covariant channel. Dependency versions, figure data and run instructions are supplied.

Declaration of generative AI and AI-assisted technologies in the writing process

An OpenAI language-model assistant was used to help restructure and edit the manuscript, check algebraic reductions, compare selected references, and develop and run numerical verification code. The author is responsible for reviewing and approving the final scientific content and references before submission.

References

[1] E. Knill and R. Laflamme, Theory of quantum error-correcting codes, Phys. Rev. A 55 (1997) 900, doi:10.1103/PhysRevA.55.900 [quant-ph/9604034].

[2] E. Knill, R. Laflamme and L. Viola, Theory of quantum error correction for general noise, Phys. Rev. Lett. 84 (2000) 2525, doi:10.1103/PhysRevLett.84.2525 [quant-ph/9908066].

[3] D.A. Lidar, I.L. Chuang and K.B. Whaley, Decoherence-free subspaces for quantum computation, Phys. Rev. Lett. 81 (1998) 2594, doi:10.1103/PhysRevLett.81.2594 [quant-ph/9807004].

[4] R. Raussendorf and H.J. Briegel, A one-way quantum computer, Phys. Rev. Lett. 86 (2001) 5188, doi:10.1103/PhysRevLett.86.5188.

[5] S.-H. Li, H.-C. Yang, R.-Y. Xu and S.-M. Wu, Does relativistic motion really freeze initially maximal entanglement?, JHEP 03 (2026) 218, doi:10.1007/JHEP03(2026)218 [arXiv:2601.02976].

[6] Y.S. Weinstein and G. Gilbert, Storing small photonic cluster states in a dephasing environment, J. Mod. Opt. 57 (2010) 1961–1964, doi:10.1080/09500340.2010.495460 [arXiv:1006.3192].

[7] Y.S. Weinstein, J. Feldman, J. Robins, J. Zukus and G. Gilbert, Superoperator analysis of entanglement in a four-qubit cluster state, Phys. Rev. A 85 (2012) 032324, doi:10.1103/PhysRevA.85.032324 [arXiv:1011.6336].

[8] K. Macieszczak, M. Guţă, I. Lesanovsky and J.P. Garrahan, Towards a theory of metastability in open quantum dynamics, Phys. Rev. Lett. 116 (2016) 240404, doi:10.1103/PhysRevLett.116.240404 [arXiv:1512.05801].

[9] W.G. Unruh, Notes on black-hole evaporation, Phys. Rev. D 14 (1976) 870, doi:10.1103/PhysRevD.14.870.

[10] J. Hu and H. Yu, Entanglement dynamics for uniformly accelerated two-level atoms, Phys. Rev. A 91 (2015) 012327, doi:10.1103/PhysRevA.91.012327 [arXiv:1501.03321].

[11] E.B. Davies, Markovian master equations, Commun. Math. Phys. 39 (1974) 91, doi:10.1007/BF01608389.

[12] S. Schlicht, Considerations on the Unruh effect: causality and regularization, Class. Quant. Grav. 21 (2004) 4647, doi:10.1088/0264-9381/21/19/011 [gr-qc/0306022].

[13] G. Kaplanek and C.P. Burgess, Hot accelerated qubits: decoherence, thermalization, secular growth and reliable late-time predictions, JHEP 03 (2020) 008, doi:10.1007/JHEP03(2020)008 [arXiv:1912.12951].

[14] S.-W. Han, W. Chen, L. Chen, Z. Ouyang and J. Feng, Reliable quantum master equation of the Unruh–DeWitt detector, Phys. Rev. D 112 (2025) 085014, doi:10.1103/c9wz-lm5m [arXiv:2502.06411].

[15] G. Vidal and R.F. Werner, A computable measure of entanglement, Phys. Rev. A 65 (2002) 032314, doi:10.1103/PhysRevA.65.032314 [quant-ph/0102117].

[16] V. Gorini, A. Kossakowski and E.C.G. Sudarshan, Completely positive dynamical semigroups of N-level systems, J. Math. Phys. 17 (1976) 821, doi:10.1063/1.522979.

[17] G. Lindblad, On the generators of quantum dynamical semigroups, Commun. Math. Phys. 48 (1976) 119, doi:10.1007/BF01608499.

[18] R.H. Dicke, Coherence in spontaneous radiation processes, Phys. Rev. 93 (1954) 99, doi:10.1103/PhysRev.93.99.

[19] M. Horodecki, P. Horodecki and R. Horodecki, General teleportation channel, singlet fraction, and quasidistillation, Phys. Rev. A 60 (1999) 1888, doi:10.1103/PhysRevA.60.1888 [quant-ph/9807091].